VIYAL TECH
INNOVATIONS

PROPOSED SOFTWARE SOLUTION

AI-Powered Academic Admission Assistance, Visitor Journey & Queue Management System

A unified digital admission experience for parents, students and college staff

Prepared by: ViyalTech Innovations

Customer segment: Colleges, Universities & Academic Institutions

Document: Client-Facing Project Explanation

Version: 1.0 | **Date:** September 2026

Table of Contents

1. Project Introduction
2. Core Idea of the System
3. Complete Visitor Journey
4. QR Code System
5. WhatsApp Communication
6. AI Admission Assistant
7. College Knowledge Base
8. Unknown Requirement / Counselling & Help Desk
9. Receptionist System
10. Digital Queue Management
11. Multi-Department Support
12. Multi-Stage Workflow Engine
13. Automatic Next-Stage Allocation
14. Staff Dashboard
15. Admin Dashboard
16. Database Technology
17. Backend Technology
18. Frontend Technology
19. Authentication & Security
20. Webhook Technology
21. Notification System
22. Complete Realistic Example
23. Complete System Architecture
24. Technology Stack Table
25. Benefits to the College
26. Future Customization
27. Future Enhancements
28. Final Summary

During every admission season, a college receives hundreds of parents and students. Most arrive not knowing the answers to a few very simple but critical questions:

- **Whom should I meet?** They often do not know which staff member handles their requirement.
- **Which department do I approach?** Computer Science, Electronics, Mechanical, Management — it is often unclear.
- **Where is the office, counter or room?** Campus layouts are unfamiliar to first-time visitors.
- **How many people will I visit during admission?** Counselling, verification, accounts and registration are usually separate counters.
- **How long will I wait?** Physical queues give no indication of position or time.
- **What is the next step after a counter?** After one stage is finished, the parent must look for the next one with very little guidance.
- **Every department follows a different process.** One uniform instruction sheet does not fit all admission procedures.

Today this situation leads to repeated questions to the reception desk, long physical queues, confused visitors, and heavy workload on staff. ViyalTech Innovations proposes a single integrated platform that solves all of these together:

```
QR ACCESS + WHATSAPP + AI ASSISTANT + RECEPTIONIST REGISTRATION
  + VISITOR ID + DIGITAL QUEUES + CONFIGURABLE WORKFLOWS
  + STAFF MANAGEMENT + AUTOMATIC NEXT-STAGE ALLOCATION
  + WHATSAPP NOTIFICATIONS
```

The platform guides every parent from the moment they arrive until their admission is fully completed, while keeping the college staff in complete control of the process at every step.

2. Core Idea of the System

This is **not simply a chatbot**. The proposed system is a: **"Digital Visitor Journey, Queue and Workflow Management System with an AI Admission Assistant."**

Every visitor is represented by **one permanent Visitor ID** for their entire visit. Regardless of how many counters, queues or stages they pass through, the same identity follows them:

```
Visitor ID : V121
Name       : Raj Kumar
Purpose    : Admission
Department : ECE

V121 → Counselling → Eligibility Verification → Document Verification
     → Accounts → Admission Confirmation → Registration → Completed
```

At each stage the visitor receives a fresh queue token (for example `ECE-COUNS-124`, then `ECE-DOC-…`), but the permanent Visitor ID stays the same. This one-ID concept gives the college:

- **Full traceability** — every stage and action can be traced to one visitor,
- **A clear audit trail** — who processed the visitor, when, and at which stage,
- **Accurate statistics** — admissions per department, stage times and completions, and
- **Consistent communication** — WhatsApp messages always speak about the same visitor.

3. Complete Visitor Journey

The system is best explained as one continuous journey. The complete flow is:

```
PARENT / STUDENT ARRIVES
        |
        ▼
QR CODE  →  opens the college's WhatsApp assistant entry point
        |
        ▼
WHATSAPP  →  parent chats with the AI admission assistant
        |
        ▼
AI ADMISSION ASSISTANT  →  answers early questions, explains the process
        |
        ▼
```

```
VISITOR ID  →  e.g. VI21 created
        │
        ▼
PURPOSE / DEPARTMENT IDENTIFICATION  →  known dept, or Counselling / Help Desk
        │
        ▼
INITIAL QUEUE  →  visitor placed in the right first-stage queue
        │
        ▼
STAFF PROCESSING  →  assigned staff member handles the stage
        │
        ▼
STAGE COMPLETION  →  staff marks the stage complete
        │
        ▼
AUTOMATIC NEXT STAGE  →  system finds and assigns the next step
        │
        ▼
NEXT QUEUE  →  visitor automatically added to the next queue
        │
        ▼
WHATSAPP NOTIFICATION  →  "next step is …, please wait for your turn"
        │
        ▼
NEXT STAFF / LOCATION  →  visitor walks to the assigned room / counter
        │
        ▼
FINAL COMPLETION  →  all stages done, admission completed
```

Step-by-step experience: a parent scans the QR at the entrance and starts chatting with the AI on WhatsApp immediately. The AI answers their early questions (documents, fees, hostel, timings). At the reception they are formally registered and given a Visitor ID. From that moment, every stage — queueing, being called, being served, and moving to the next counter — is managed by the system and communicated to them on WhatsApp. Staff see exactly who is waiting for them. The parent never has to guess where to go next.

4. QR Code System

The QR code is the visitor's simplest gateway into the system. It is placed in easy-to-see locations:

- college main entrance,
- reception desk,
- admission block,
- help desk, and
- posters and signboards across the campus.

```
QR SCAN
   │
   ▼
Opens the college's WhatsApp communication entry point
   │
   ▼
Parent sends a message (often pre-filled, e.g. "Hi, I came for admission")
   │
   ▼
The AI admission assistant replies and becomes available
```

The moment the QR is scanned, WhatsApp opens a conversation with the college's admission assistant. No mobile application needs to be downloaded or installed, and no phone number has to be remembered or typed. The visitor is connected to the system in seconds.

QR access benefits the college too: it is cheap to print, reusable, and carries the parent to a communication channel most people already use every day.

5. WhatsApp Communication

Communication between the parent and the college runs over **real WhatsApp**, powered by the official **Meta WhatsApp Business Platform / Cloud API**. WhatsApp is the channel the system uses to talk to the visitor.

```
      ▼
Meta WhatsApp Cloud API
      |
      ▼
Webhook  (secure callback into the college backend)
      |
      ▼
Backend  (AI + Queue + Workflow + Database)
      |
      ▼
Backend  →  WhatsApp API  →  Parent WhatsApp
```

Parents can interact with the system entirely from WhatsApp:

- ask admission and college questions,
- receive basic college information,
- check their live queue position,
- view their current stage and the next step,
- get location and room guidance,
- receive turn-call notifications, and
- receive stage-completion and process updates.

WhatsApp is not an afterthought — the system is designed around it as the primary visitor-facing channel. Because parents already use WhatsApp daily, there is no learning curve, and the college does not need to build or distribute a separate parent application.

6. AI Admission Assistant

The AI assistant is the "face" of the system in conversation. It is built on a **local AI architecture** using **Ollama**, a tool that runs a **local large language model** on the college's own hardware.

```
Parent message
      |
      ▼
WhatsApp
      |
      ▼
Backend
      |
      ▼
AI Service
      |
      ▼
Ollama  →  Local Language Model
      |
      ▼
Generated response
      |
      ▼
WhatsApp → Parent
```

The AI understands natural questions and answers them in simple language, for example:

> **Parent:** "What documents are required for ECE admission?"
> **AI:** "For ECE admission you will need Class 10 and 12 mark sheets, certificates, transfer certificate and photographs — brought in original."

> **Parent:** "Where is the admission office?"
> **AI:** "The admission office is in the Administrative Block, first floor, Room A-102. The reception can guide you."

It also answers: "What courses are available?", "What are the admission timings?", "Is hostel available?", "What is the fee?", "Where should I go next?" and "How many people are before me in the queue?"

Clear separation of responsibilities

Database	Stores the authoritative college information and knowledge base.
Workflow Engine	Determines the visitor's actual process and next stage.
Queue Engine	Determines the real queue state and positions.
Staff	Control and complete the real-world admission stages.

An important design principle: **the AI never changes the queue or the workflow on its own**. It can report, explain and guide, but official visitor state changes only through authorized staff and receptionist actions. This keeps the process trustworthy and auditable.

7. College Knowledge Base

Rather than letting the AI invent answers, the system maintains a **college knowledge base** — a structured store of the college's own information, managed by the college itself:

- college details and contact information,
- departments and courses,
- admission information, eligibility and fees,
- required documents,
- hostel and transport,
- timings,
- location, rooms and counters, and
- frequently asked questions.

When a parent asks a question, the AI builds its answer from this configured knowledge base. The information the parent receives is therefore the college's own up-to-date information, not a generic or guessed reply.

Parent question	Source in the knowledge base
"What documents are required?"	Departmental / admission FAQ entry
"What is the fee?"	Fees entry
"Is hostel available?"	Hostel entry
"What are the timings?"	Timings entry
"Where is the admission office?"	Location entry
"What is my queue position?"	*Live queue data* (not the knowledge base)

The knowledge base is maintained through the Admin dashboard (Section 15), so information stays current without any technical work.

8. Unknown Requirement / Counselling & Help Desk

Many visitors genuinely do not know which department or course they need. The system is designed to handle this gracefully instead of forcing a guess.

> **Parent:** "I don't know which department I need."
> **AI:** "No problem — our counselling / help desk team will guide you based on your interests and marks. You will not be assigned to any department automatically."

- When a visitor is unsure, the receptionist registers them under **"Needs Counselling / Help Desk"**.
- The system routes them into the dedicated **Counselling / Help Desk** workflow and queue, not into any random department.
- Help desk staff interview the visitor, understand their requirement, and direct them to the correct department and process.

Why this matters: an unsure visitor placed into the wrong department creates a wrong journey, wastes staff time and upsets the visitor. Help-desk routing keeps the experience accurate and respectful, and it is handled by the same queue and workflow machinery — no special cases.

9. Receptionist System

```
NEW VISITOR REGISTRATION
----------------------------------------
 Name      : Raj Kumar
 WhatsApp/Mobile : 9876543210
 Purpose   : Admission
 Department: ECE            (or "Needs Counselling / Help Desk")
 Remarks   : Walk-in parent
----------------------------------------
 [REGISTER VISITOR]  →  creates Visitor ID: V121
```

With one click the system:

- creates the permanent Visitor ID (V121, V122, ...),
- selects the right admission workflow (department or help desk),
- places the visitor into the correct first-stage queue with a token,
- records the stage history, and
- sends a welcome WhatsApp message with the Visitor ID, stage, queue token and instructions.

This single registration action connects the *physical visitor standing at the counter* with their complete *digital journey* — from there on, the AI, the queues, the staff dashboards and WhatsApp notifications all recognize the same visitor.

10. Digital Queue Management

The system replaces physical standing-in-line with **real digital queues**. Every admission stage has its own queue with known numbers and positions. A visitor always knows where they stand.

Status	Meaning
WAITING	Visitor is in the queue and waiting for their turn.
CALLED	Staff pressed "Call Next"; the visitor's number has been announced.
SERVING	Staff is currently attending to the visitor.
COMPLETED	The stage is finished; the visitor may be moved to the next queue automatically.
HOLD	Processing paused for this visitor (e.g. missing a document); others are served first.
SKIPPED	Visitor left this queue run (e.g. rescheduled, asked to come back later).
CANCELLED	Visitor record cancelled (no-show or left the college).

What staff and the system can do with a queue:

- show the queue number, current position and number of people ahead,
- show who is currently being served,
- **call the next visitor** — the earliest waiting visitor is summoned and notified on WhatsApp,
- **hold** or **skip** a visitor with a recorded reason,
- **reassign** a visitor to a different stage or queue, and
- **complete** a stage and automatically prepare the next one.

A realistic example:

```
ECE Counselling Queue          Document Verification Queue
E-101  – Serving                DV-41 – Serving
E-102  – Waiting                DV-42 – Waiting
E-103  – Waiting                DV-43 – Waiting (this is V121's next number)
```

Visitors no longer need to crowd around counters. They wait comfortably, watch their number on WhatsApp, and walk to the counter only when called.

11. Multi-Department Support

The system serves an entire college with many departments — for example **CSE, ECE, EEE, Mechanical, MBA**, plus the shared **Help Desk**. Each department has its own queues, staff and process.

The system deliberately does **not** assume every department follows the same procedure. A Computer Science admission may include an aptitude or counselling step; a Management admission may include a group discussion round; an

12. Multi-Stage Workflow Engine

The **workflow engine** is the core of the system. A **workflow** is a defined chain of **stages**; each stage knows its department, staff, room or counter, queue and order. Different departments have different workflows.

```
ECE Admission:
  Counselling → Eligibility Verification → Document Verification
  → Accounts → Admission Confirmation → Registration

CSE / EEE / Mechanical / MBA: their own, separately configurable chains
Counselling / Help Desk: a single help-desk stage for unsure visitors
```

An administrator can configure for each workflow:

- the workflow itself and the department it belongs to,
- the list of stages and their exact order,
- the staff member assigned to each stage,
- the room / counter used by each stage,
- the queue behind each stage, and
- whether a stage is active or inactive.

Because workflows are *configuration data* rather than hard-coded logic, changing the admission process is a simple administrative change — no software rebuild needed.

13. Automatic Next-Stage Allocation

This is the most important automation in the system. Consider visitor **V121 at Counselling**. When the staff member completes the Counselling stage, the system performs seven actions automatically:

```
Staff clicks COMPLETE for V121 (Counselling)
        |
        ▼
1. Marks Counselling complete
2. Finds the next workflow stage  →  Document Verification
3. Identifies the next required queue
4. Places V121 into that queue (new token)
5. Updates V121's current stage
6. Identifies the appropriate staff and location
7. Sends the WhatsApp notification
```

Example notification:

> **WhatsApp → V121:**
> "Counselling completed. Your next step is Document Verification.
> You have been added to the Document Verification queue. Please wait for your turn."

This chain repeats automatically after every stage until the visitor completes the entire workflow. The parent is guided from counter to counter without ever asking "what happens now?".

14. Staff Dashboard

Each staff member sees only the queues they are responsible for. The staff dashboard shows:

- current queue and its size,
- visitors waiting with their tokens and positions,
- the visitor currently being served,
- visitor details (name, phone, purpose, department, remarks),
- the current stage, room/counter and department, and
- the live status of every entry.

Staff actions and their effect:

Action	Effect

HOLD	Pauses the visitor (with a reason) so others can be served first.
SKIP	Removes the visitor from this queue run (with a reason).
REASSIGN	Moves the visitor to another stage/queue when appropriate.
VIEW	Opens the full visitor profile and history.
COMPLETE	Finishes the stage and triggers automatic next-stage allocation (Section 13).

The dashboard is simple, fast and focused — a staff member never sees queues that do not belong to them.

15. Admin Dashboard

The Admin dashboard gives the college full control of its configuration. The administrator manages:

- departments,
- staff accounts and assignments,
- rooms and counters,
- workflows and workflow stages,
- FAQs and knowledge-base entries,
- college information (fees, hostel, transport, timings, locations),
- queue settings, and
- visitor records and reassignments.

With these tools the college can change its whole admission design — add a stage, change a counter, update an answer in the knowledge base — entirely from the dashboard. This is what makes the system **configurable rather than hard-coded**: technology adapts to the college's process, not the other way around.

16. Database Technology

The system stores everything in a database using **SQLite** with the **SQLAlchemy** object-relational layer. A database is essential because every queue, visitor, workflow and message must be persistent, consistent and queryable in real time.

```
Users   ↔ Staff   →  Departments
Visitors → Queue Entries → Queues → Workflow Stages → Workflows → Departments
Visitors → Visitor Stage History (audit trail)
Visitors → Notifications / Chat Messages
Workflow Stages → Rooms · Staff · Queues
FAQs · College Information → knowledge base for the AI
```

The major entities in plain language:

- **Users** — logins for admin, reception and staff,
- **Visitors** — every registered visitor with their permanent Visitor ID,
- **Departments & Staff & Rooms** — the college's structure: who works where,
- **Workflows & Workflow Stages** — each department's admission process (ordered stages),
- **Queues & Queue Entries** — one real queue per stage and every visitor's token and status,
- **Visitor Stage History** — the complete audit trail of a visitor's journey,
- **FAQs & College Information** — the knowledge base the AI answers from,
- **Notifications & Chat Messages** — the communication log.

The relationships above allow the system to answer questions such as "what is V121's full journey?", "who is ahead in this queue?" and "how many visitors are waiting at every stage?" instantly and accurately.

17. Backend Technology

The backend is built with **Python** and **FastAPI**, which act as the central hub of the entire system. The backend is responsible for:

- **Authentication** — logging in users and issuing secure tokens,
- **Visitor management** — registration, search, status, history,
- **Queue management** — creating queues, assigning tokens, computing positions, calling next,
- **Workflow management** — stages, ordering, next-stage logic,

- **Notifications** — generating and sending WhatsApp notifications,
- **Database operations** — all reads and writes through SQLAlchemy, and
- **Role permissions** — enforcing what each role may do.

FastAPI is well suited because it is fast, modern, and well-documented, and it cleanly separates routes, business logic (services) and data access — keeping the system maintainable as the college's needs grow.

18. Frontend Technology

The frontend is built with **React**, bundled by **Vite**, and styled with **Tailwind CSS**. It provides the complete, modern web interface used by the college team:

- **Admin dashboard** — configuration and statistics,
- **Receptionist dashboard** — registration and queue overview,
- **Staff dashboard** — live queue console and actions,
- visitor and queue management screens,
- workflow management screens,
- data-entry forms and search/filtering,
- tables with clear status indicators and badges, and
- a responsive design that works well on desktop, laptop, tablet and phone.

The user interface is role-aware: each role sees only the menus and actions it is permitted to use. Combined with the backend security (Section 19), this gives a safe, uncluttered experience for every role.

19. Authentication & Security

The system is protected end-to-end with three complementary mechanisms:

- **JWT authentication** — after a successful login the user receives a secure token that authorizes every subsequent request,
- **Password hashing** — passwords are never stored in plain text, and
- **Role-based access control (RBAC)** — each role may only do what it is allowed to.

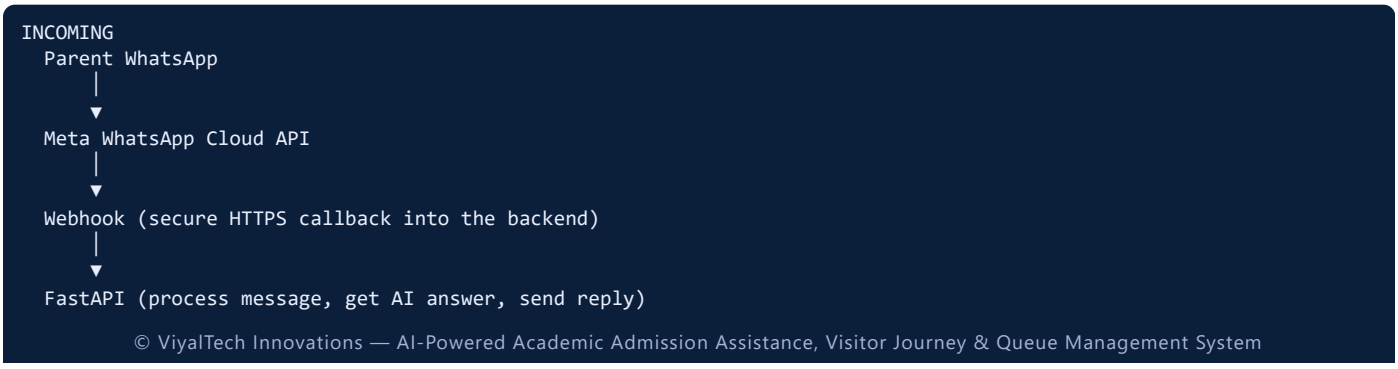
Role	Permissions
ADMIN	Full control: departments, staff, rooms, workflows, stages, knowledge base, statistics, all visitor records.
RECEPTIONIST	Visitor registration, search, active visitors, all queues, reassignment, cancellation.
STAFF	Only their assigned queues/stages and processing actions (call, hold, skip, reassign, complete).

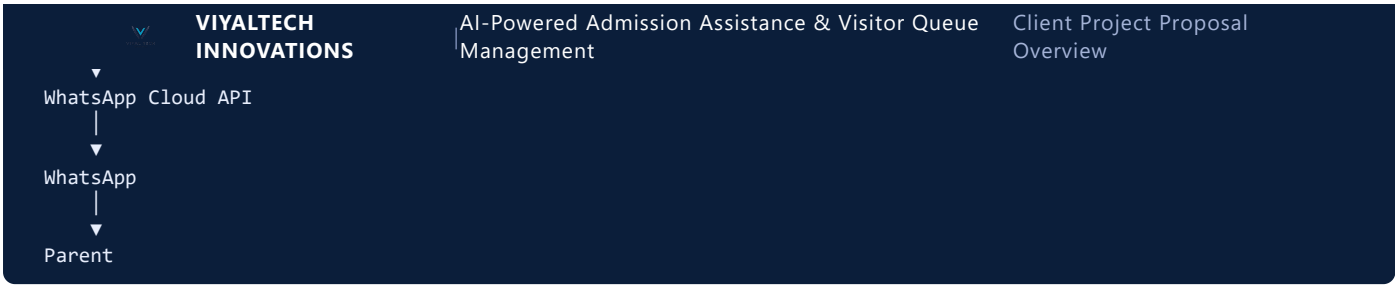
Configuration and secrets follow security best practice:

- all API credentials, WhatsApp tokens and secrets are kept in environment variables (a `.env` file),
- the repository includes only a template (`.env.example`) with variable names — never values,
- the JWT signing secret, WhatsApp access tokens and phone-number ID are protected and never hard-coded, and
- production deployment will use HTTPS so all traffic is encrypted.

20. Webhook Technology

A **webhook** is a secure callback: when an event happens on WhatsApp, Meta's platform calls a specific URL at the college's backend to deliver the message. The backend then processes it and sends an answer back.





Webhook communication is necessary because WhatsApp (via Meta) can push messages to the college server in real time — the college does not need to repeatedly ask WhatsApp "are there new messages?". This makes the assistant genuinely conversational and immediate.

21. Notification System

The system automatically sends WhatsApp notifications whenever something important happens in the visitor's journey:

- **Visitor registration** — welcome, Visitor ID, stage and token,
- **Queue position** — on request, the live position and people ahead,
- **Turn approaching / called** — "Your turn is now. Please proceed to Room…",
- **Next stage** — "… your next step is Document Verification",
- **Room / location and staff assignment** — where and whom to meet,
- **Stage completion** — confirmation of the finished stage, and
- **Final completion** — "All your admission stages are complete. Congratulations!".

Notifications are generated by the backend *at the moment of the actual event* — they always reflect the true official state of the visit, never a guess. The visitor always knows exactly where they are and what comes next.

22. Complete Realistic Example — Raj Kumar (V121)

The following example shows the entire system working for one parent, Raj Kumar, seeking ECE admission. WhatsApp messages shown are examples of what the parent would receive.

```
ARRIVAL → scans QR → chats with AI → asks about documents and fees
    │
RECEPTION → registers: Raj Kumar · 9876543210 · Admission · ECE
    │   → "Welcome Raj! Your Visitor ID is V121. You are in the
    │     Counselling queue. Your number is ECE-COUNS-124."
    ▼
V121 generated → placed in ECE Counselling queue (WAITING)
    │
AI during the wait: answers "How many people before me?"
    │   → "V121: 2 people ahead of you at Counselling."
    ▼
Staff (Counsellor) → CALL NEXT
    │   → SMS/WA: "Your turn! ECE-COUNS-124. Proceed to Counselling
    │     Room (R101). Meet: Mrs. Priya."
    ▼
Counselling COMPLETED
    │   → WA: "Counselling completed. Next step: Document Verification.
    │     You have been added to that queue. Please wait."
    ▼
Document Verification queue → staff COMPLETES
    │   → WA: "Document Verification completed. Next: Accounts."
    ▼
Accounts queue → staff COMPLETES
    │   → WA: "Accounts completed. Next: Admission Confirmation."
    ▼
Admission Confirmation → staff COMPLETES
    │   → WA: "Confirmed. Next: Registration."
    ▼
Registration (final stage) COMPLETED
    │   → WA: "All your admission stages are complete. Congratulations!"
    ▼
V121 → COMPLETED
```

At every step Raj knew where to go, whom to meet and what came next — without asking reception once. His one Visitor ID, V121, represented his whole journey.



How all components communicate: the parent interacts through WhatsApp. Meta forwards incoming messages to the backend webhook. The backend asks the AI service — which uses Ollama and the local language model plus the college knowledge base and the visitor's live state — to produce an answer, and replies via the WhatsApp API. All queues, workflows, visitors and messages live in the database. The college team manages everything through the role-based React frontend, which talks to the backend over protected APIs. In one sentence: *WhatsApp talks to the parent, the backend orchestrates the journey, the database records everything, and staff stay in control.*

24. Technology Stack Table

Layer	Technology	Purpose
WhatsApp communication	Meta WhatsApp Business Platform / Cloud API	Real parent messaging, inbound + outbound
QR-based access	QR codes linked to the WhatsApp assistant	Instant entry point for visitors
Backend	Python + FastAPI	Central API layer, webhook, orchestration
AI	Ollama + local large language model	Conversational admission assistant
Database	SQLite + SQLAlchemy	Persistent relational storage (14 major entities)
Frontend	React + Vite	Admin/Reception/Staff web dashboards
UI styling	Tailwind CSS	Modern responsive interface
Authentication	JWT + password hashing	Secure login and sessions
Communication	Webhook (HTTPS callback)	Real-time inbound WhatsApp messages
Queue management	Custom backend queue engine	Real queues, positions, call-next logic
Workflow	Configurable workflow engine	Department-specific multi-stage processes

25. Benefits to the College

Benefit	Impact

Better visitor experience	Guided, respectful, low-stress admission visit.
Less physical queue congestion	Visitors wait comfortably and come when called.
Centralized visitor tracking	One system, one Visitor ID, complete visibility.
Better staff coordination	Each staff member sees exactly their queue and their work.
Automated visitor guidance	Automatic next-stage flows remove guesswork.
Department-specific processes	Each department follows its own correct workflow.
Clear admission journey	The full path is visible to staff, admin and visitor.
Real-time queue visibility	Positions and waiting lists updated live.
Reduced repetitive questions	The AI answers routine queries on WhatsApp 24×7.
Better admission-day organization	Predictable flow, fewer bottlenecks, calmer counters.
Digital records	Complete documented audit of every visitor journey.

26. Future Customization

The system is designed to be customized to any college. Because departments, courses, stages, staff, rooms, counters, FAQs and policies are configuration rather than code, the college can shape the system to its own identity without any programming:

- departments and courses offered by the college,
- admission stages and their order per department,
- staff, rooms and counters used in each stage,
- FAQs and the entire college knowledge base,
- college policies, timings and contact details, and
- workflow rules for each admission track.

ViyalTech Innovations would work with the college during setup to configure all of this to match its real admission process.

27. Future Enhancements

Beyond the core system described here, the following capabilities can be added over time. They are listed as future possibilities and are *not part of the core system* initially:

- multilingual AI — answers in regional languages,
- voice-based assistance — voice messages and speech-to-text,
- document upload — parents share scanned documents via WhatsApp,
- appointment scheduling — pre-booking counselling and verification slots,
- admission analytics — dashboards and trend reports,
- mobile staff application — call/complete from a phone,
- cloud deployment — hosting on college or cloud infrastructure,
- advanced document retrieval / RAG — searching brochures and circulars, and
- additional college integrations — ERP, fee and student systems.

28. Final Summary

The proposed system creates a **single unified digital journey** for every parent and student who comes to the college for admission.

```
QR          →  instant access to the assistant
WHATSAPP    →  real communication with the visitor
AI          →  answers questions and guides the visitor
RECEPTION   →  human registration and Visitor ID
QUEUE       →  real digital queues with live positions
STAFF       →  process and complete each stage
WORKFLOW    →  department-specific multi-stage processes
AUTO NEXT   →  automatic assignment of the next stage
NEXT QUEUE  → visitor moves seamlessly to the next counter
NOTIFY      →  WhatsApp updates at every step
COMPLETED   → admission journey finished successfully
```

and complete digital records.

> **ViyalTech Innovations** proposes a complete, integrated solution: an AI-powered assistant, real WhatsApp communication, QR-based access, digital queues and configurable multi-stage workflows that can be adapted to the college's exact admission process.

— End of document · ViyalTech Innovations · AI-Powered Academic Admission Assistance, Visitor Journey & Queue Management System —